Bilipschitz Equivalence of Integer Power Word Metrics

Proof and Lean Formalization Report

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Abstract

For an integer $a \geq 2$, let d_a be the word metric on $\mathbb{Z}$ generated by $\{\pm a^k : k \in \mathbb{N}\}$. We give a proof that all these metric spaces admit bilipschitz bijections. The central construction is a relative Hamming recoding that preserves two distinguished symbols, including their positions, and commutes with changes between those symbols. This property turns an arbitrarily long carry into the difference of two generators. The Lean implementation uses a Hamming constant of 35 and yields forward and reverse constants 564 and 321 for bases 2 and 3. A Lean formalization of the proof accompanies this report.

1 The question and precise result

Define

$$\ell_a(z) = \min\left\{ r \in \mathbb{N} : z = \sum_{j=1}^{r} \varepsilon_j a^{k_j}, \quad \varepsilon_j \in \{-1, 1\},\ k_j \in \mathbb{N} \right\}, \qquad d_a(x, y) = \ell_a(y - x).$$

The empty sum represents zero. The minimum exists because $1 = a^0$ is a generator. Nathanson [1, Section 4, Problems 1–2] asks whether $(\mathbb{Z}, d_a)$ and $(\mathbb{Z}, d_b)$ are bilipschitz equivalent, or more generally quasi-isometric, for integers $a, b \geq 2$. The case $a = 2, b = 3$ is of particular interest. The equivalence may be an arbitrary map of metric spaces; it need not be the identity map or a group homomorphism.

Theorem 1. *For every $a, b \in \mathbb{N}$ with $a, b \geq 2$, there exists a bijection $f : \mathbb{Z} \to \mathbb{Z}$ such that, for all $x, y \in \mathbb{Z}$,*

$$d_b(f(x), f(y)) \leq a\big(2 + 35(b^2 - 1)\big) d_a(x, y), \tag{1}$$

$$d_a(x, y) \leq b\big(2 + 35(a^2 - 1)\big) d_b(f(x), f(y)). \tag{2}$$

Consequently these spaces are bilipschitz equivalent, and hence quasi-isometric.

Corollary 2. *There exists a bijection $f : \mathbb{Z} \to \mathbb{Z}$ satisfying*

$$\frac{1}{321} d_2(x, y) \leq d_3(f(x), f(y)) \leq 564 d_2(x, y) \quad (x, y \in \mathbb{Z}).$$

In particular, a common bilipschitz constant is 564.

The formal theorem is an existence theorem in classical mathematics. It makes no assertion that the chosen bijection is computable, additive, affine, finite-state, or continuous for an adic topology. The constants above are those proved in the formal development.

2 Relative recoding of finite words

Let A be a countable alphabet with distinguished element 0 and at least two elements. Add two new symbols P, M, and let $\mathcal{S}(A)$ be the set of finite-support functions $\mathbb{N} \to A \sqcup \{P, M\}$, based at 0. Equip it with Hamming distance. Write

$$F(x) = \{j : x_j \in \{P, M\}\}$$

for the finite frozen set. A flip on a finite set of coordinates interchanges P and M there and fixes all ordinary letters.

Lemma 3 (Relative Hamming equivalence)**.** *For any two such alphabets A, B, there is a bijection $T : \mathcal{S}(A) \to \mathcal{S}(B)$ with both T and T^{-1} 35-Lipschitz for Hamming distance. At every coordinate it preserves whether the letter is P, M, or ordinary. It also commutes with every finite coordinate flip.*

Ordinary Hamming equivalence over finite alphabets is established in Domat–Hoganson–Lyman [2, Section 4.2]. Their absorption and infinite partition method motivates the construction below. The additional frozen set and flip conditions are established explicitly here; the ordinary Hamming theorem alone does not supply the carry argument.

2.1 An absorption matching on an infinite set

Let E be a countably infinite set and let $A^{(E)}$ denote its finitely supported A-valued functions. For a pointed countable alphabet B, relate $x \in A^{(E)}$ to $(b, z) \in B \times A^{(E)}$ when x, z differ in at most one coordinate. Every vertex on each side has infinitely many neighbors: choose a fixed nonzero letter of A and insert it outside the finite support. Enumerating either countable side and choosing a fresh neighbor gives injections in both directions which respect this same relation. The relation-preserving Schröder–Bernstein theorem then gives a bijection

$$\psi_E : A^{(E)} \longrightarrow B \times A^{(E)}, \qquad \psi_E(x) = (b, z), \qquad \mathrm{Ham}(x, z) \leq 1.$$

Reserve the edge $0 \leftrightarrow (0, 0)$ before the matching construction to ensure $\psi_E(0) = (0, 0)$. The inverse has the same displacement property. Choose ψ_E as a function of the set E alone. This last condition is essential when comparing different frozen sets.

2.2 Assembly with a variable frozen set

Fix a partition $\mathbb{N} = \bigsqcup_{i \in \mathbb{N}} B_i$ into infinite sets; formally, B_i is the fiber of the first component of the standard pairing inverse. For a frozen set F, put $E_i = B_i \setminus F$, which remains infinite. If $i \notin F$, apply ψ_{E_i} to the ordinary input data on E_i, retaining its modified A-word and storing its extra B-letter at coordinate i. If $i \in F$, use the identity on E_i and store no extra letter. Copy every frozen label unchanged. At each editable coordinate, pair the retained A-letter with its stored B-letter. This defines

$$\Phi_{A,B} : \mathcal{S}(A) \longrightarrow \mathcal{S}(A \times B).$$

It is bijective: from an output, recover F, split its ordinary pairs, and apply the corresponding inverse matching in each active block. Only finitely many input blocks are nonzero, and every local matching fixes zero; thus both directions preserve finite support. The construction depends on the frozen positions but not on the values P versus M. It therefore commutes with all flips.

2.3 Uniform control across different frozen sets

Here is the estimate that makes the construction global. Given two input words, let D be their difference set, $d = |D|$, and let $\pi(j) = i$ for $j \in B_i$. Only indices in

$$I = \pi(D) \cup D, \qquad |I| \le 2d,$$

can have changed block input, editable set, or active status. For $i \notin I$, both words have the same editable set and the same data on B_i, and coordinate i agrees. They therefore use exactly the same local matching and produce identical retained data on B_i and the same extra letter at coordinate i.

Within an affected block the retained A-word differs from the original at at most one position for each of the two inputs. The difference count of the retained first components is consequently at most $d + 2|I| \le 5d$. The extra components differ at at most $|I| \le 2d$ coordinates. Hence $\Phi_{A,B}$ is 7-Lipschitz. For the inverse, let D instead be the difference set of the two input words in $\mathcal{S}(A \times B)$. A changed extra B-letter at coordinate i places i in D, so its affected inverse block is already counted in $I = \pi(D) \cup D$. Projecting to the A-components and retaining frozen labels gives at most d differences; each inverse matching changes this projected word at at most one position per affected block, for each input. The inverse therefore has at most $d + 2|I| \le 5d$ differences, with no separate extra-letter term. Thus $\Phi_{A,B}^{-1}$ is 5-Lipschitz. This reasoning includes changes in membership of F, rather than only pairs of words with one fixed frozen set.

Apply the construction with the alphabets reversed and use the coordinate swap $A \times B \cong B \times A$. Then

$$T = \Phi_{B,A}^{-1} \circ \mathrm{swap} \circ \Phi_{A,B}$$

has constant $5 \cdot 7 = 35$ in both directions and retains all frozen and flip properties. This proves Lemma 3.

3 Signed expansions and the carry estimate

For $A \ge 4$, set $D_A = \{-1, 0, 1, \dots, A-2\}$. Every integer has a unique finite expansion

$$n = \sum_{i \ge 0} d_i A^i, \qquad d_i \in D_A. \tag{3}$$

Indeed, the next quotient is $q = \lfloor (n+1)/A \rfloor$ and the remainder $d = n - Aq$ belongs to D_A. For $n \ne 0$, $|q| < |n|$, so the process terminates. For uniqueness, the least differing digit in two finite expansions would give a nonzero difference divisible by A with absolute value at most $A - 1$, which is impossible.

Identify $A - 2$ with P and -1 with M; the other $A - 2$ digits form the ordinary alphabet. Its size is at least two. Let $E_A : \mathbb{Z} \to \mathcal{S}(\{0, \dots, A-3\})$ be the resulting expansion bijection. Given the relative recoding T for bases $A, B \ge 4$, define

$$f_{A,B} = E_B^{-1} \circ T \circ E_A.$$

Fix a generator edge $n \mapsto n + A^k$. Starting at position k, let t be the first digit below $A - 2$; it exists because the word is eventually zero. The addition changes the digits $k, \dots, t-1$ from P to M and increments only the digit at t. This follows from $(A-2) + 1 = -1 + A$ and applies to negative integers as well.

First perform just those frozen flips, producing a word u. Its recoded image differs from $T(E_A(n))$ by precisely the same flips. Decoding in base B, the numerical change is

$$-(B-1)\sum_{i=k}^{t-1} B^i = B^k - B^t. \tag{4}$$

Its B-word length is at most two, including the case $t = k$, when the sum is empty. The word u differs from $E_A(n + A^k)$ at only one position, so their recoded words differ at at most 35 positions. A change of one target digit has coefficient of absolute value at most $B - 1$, costing at most $B - 1$ signed B-generators. Therefore

$$d_B\big(f_{A,B}(n + A^k), f_{A,B}(n)\big) \le 2 + 35(B-1).$$

Reverse generator edges follow by symmetry. Summing over an actual shortest signed-power word yields

$$\begin{aligned} d_B(f_{A,B}(x), f_{A,B}(y)) &\le [2 + 35(B-1)]d_A(x,y), \\ d_A(x,y) &\le [2 + 35(A-1)]d_B(f_{A,B}(x), f_{A,B}(y)). \end{aligned}$$

The second inequality is the same argument for T^{-1}, which also commutes with flips. The expansion maps themselves need no Hamming Lipschitz bound; equation (4) controls long carries.

4 Returning to all bases

For $a \ge 2$ and $z \in \mathbb{Z}$,

$$\ell_a(z) \le \ell_{a^2}(z) \le a\ell_a(z).$$

The first inequality is inclusion of generating sets. For the second, an even power of a is an a^2-generator and an odd power is a sum of a equal a^2-generators. The statement also holds for negative generators. Take $A = a^2, B = b^2$ in the preceding construction and combine these comparisons. This proves (1)–(2), including the small bases 2 and 3.

5 Formal statement and verification boundary

The Lean development defines the predicate `WordBound` inductively using exactly r signed powers. The function `wordLength` takes the least attainable r, and `wordDist a x y` is the resulting length of $y - x$. Thus the encoded distance is the original word distance, rather than a digit-weight surrogate. `PowerInt a` is a type synonym for $\mathbb{Z}$ with this distance cast to $\mathbb{R}$, equipped with a proved `MetricSpace` instance.

The main public statements are in `lean/Nathanson/Main.lean`:

Lean declaration (namespace `Nathanson`)	Mathematical content
`relativeEquiv35`	Lemma 3, both directions and flips
`all_bases_bilipschitz`	Theorem 1, integer distances
`all_bases_metric_bilipschitz`	Standard `LipschitzWith` and `AntilipschitzWith` for a metric-space equivalence
`prime_bases_metric_bilipschitz`	Specialization to any two primes
`two_three_bilipschitz`	Corollary 2, constants $564, 321$
`two_three_metric_bilipschitz`	The same constants in the metric-space interfaces

The implementation is pinned to Lean `v4.33.0-rc1` and the following mathlib commit:

`79d0395a1825a6264ad5d269e35e60537518955e`.

The proof is organized into matching, absorption, block assembly, cross-mask control, signed digits, carry transfer, and metric comparison. It includes the infinite matching and the variable-mask argument themselves; these are not imported as unproved mathematical assumptions. The local matching uses classical choice. The checked transitive logical dependencies are exactly Lean's `propext`, `Classical.choice`, and `Quot.sound`; the corresponding output is retained in `lean/build.log`.

Public repository and reproduction

The public project is available at

https://github.com/KeeneLee/integer-power-word-metrics.

The repository contains `README.md`, this report as `report.pdf`, the Lean project in `lean/`, and an MIT `LICENSE`. The 14 proof modules are in `lean/Nathanson/`; `lean-toolchain`, `lakefile.toml`, and `lake-manifest.json` inside `lean/` pin the build environment. With elan installed, run from the repository root:

```
cd lean
lake exe cache get
lake build
```

The target declarations and their `#print axioms` commands are in `lean/Nathanson/Main.lean`. The retained build log from 26 September 2026 reports `Build completed successfully (3013 jobs)`; the original build rebuilt all 14 project modules and used the pinned upstream mathlib cache. This November revision updates the public presentation and project paths, without changing the Lean proof sources or claiming a new build.

Acknowledgment

AI tools assisted with the proof and Lean formalization.

References

[1] M. B. Nathanson, *Bi-Lipschitz equivalent metrics on groups, and a problem in additive number theory*, Portugaliae Mathematica **68** (2011), 191–203. arXiv:0902.3254.

[2] G. Domat, H. Hoganson, and R. A. Lyman, *Coarse geometry of homeomorphism groups: Classifying countable Stone spaces*, version 2, 24 July 2026, Section 4.2, especially Lemma 4.7 and Theorem 4.8. arXiv:2607.01196v2.